SWABHAV techlabs
Delivering Talent with Activated Knowledge

23 October 2026

PRACTICAL MACHINE LEARNING · DAY 01 OF 18

Introduction to Practical Machine Learning

From a problem statement to a working, evaluated model

4 hours theory · 3 hours practical · Python and scikit-learn

TODAY

Seven hours, one complete workflow

Block	Time	What we cover
Theory 1	60 min	What ML is, how it works, core terminology
Theory 2	60 min	Supervised, unsupervised and other approaches
Theory 3	60 min	The ML lifecycle, applications, a fraud case study
Theory 4	60 min	Train/test splits, overfitting, evaluation metrics
Lab 1	60 min	Python environment and Iris data exploration
Lab 2	60 min	Train your first classifier (KNN)
Lab 3	60 min	Evaluate the model and experiment

LEARNING OUTCOMES

By the end of today, you can

1. Define machine learning and contrast it with rule-based programming
2. Identify features, target, observations and labels in any problem
3. Classify a problem as classification, regression or clustering
4. List the seven stages of the ML lifecycle and the purpose of each
5. Explain why models are tested on unseen data, and spot overfitting
6. Say what accuracy, precision, recall, F1, MAE, MSE and RMSE measure
7. Load, explore, split, train, predict and evaluate a model in scikit-learn

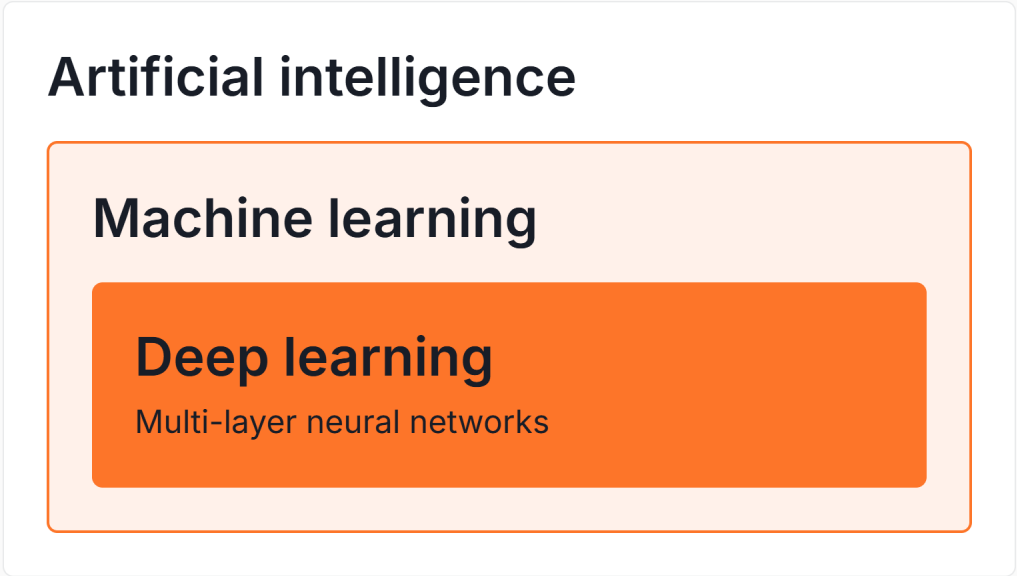
THEORY · HOUR 1 · 60 MIN

Introduction to Machine Learning

What ML is · How it works · Core terminology · Features and targets

TOPIC 1 · WHAT IS ML?

AI, machine learning and deep learning



AI — machines doing tasks that need human-like intelligence: reasoning, perception, language, decisions.

Machine learning — systems that improve at a task by learning from data, not hand-written rules.

Deep learning — ML with multi-layer neural networks; strongest on images, audio and text.

Every DL system is ML. Not every ML system is DL.

TOPIC 1 · DEFINITION

A program learns when experience improves its performance

Tom Mitchell (1997): a program learns from experience E with respect to task T and performance measure P, if its performance on T, measured by P, improves with E.

Task T

Label each incoming email as spam or not spam

Experience E

50,000 past emails already labelled by users

Performance P

Percentage of emails labelled correctly

TOPIC 1 · TWO WAYS TO BUILD SOFTWARE

Traditional programming vs machine learning

Traditional programming

Data + rules → Program → Answers

Who finds the logic? The programmer.

When the world changes: rewrite the code.

Machine learning

Data + answers → Algorithm → Rules (model)

Who finds the logic? The algorithm.

When the world changes: retrain on new data.

TOPIC 1 · WHY ML IS NEEDED

Use ML when rules alone fall short

Rules too subtle to write

Recognising a handwritten “7” in a thousand handwriting styles

Rules keep changing

Spammers change their wording every week

Pattern hidden in data

Which customers will cancel their subscription next month

Answers must be personal

Netflix recommendations for 200 million different viewers

TOPIC 1 · EXAMPLE

A rule-based spam filter breaks quickly

```
def is_spam(email):
words = ["free", "winner", "lottery"]
text = email.lower()
return any(w in text for w in words)
```

"You are a WINNER! Click here"

Rule says spam · **correct**

"Free lunch in the cafeteria today"

Rule says spam · **wrong**, it is genuine

"Fr33 l0ttery pr1ze waiting"

Rule says not spam · **wrong**, rule dodged

Exercise 1.1 · 3 min · pairs

Rules or ML? Decide, and give one reason

1

Calculate GST on an invoice

2

Recognise a face to unlock a phone

3

Convert Celsius to Fahrenheit

4

Predict tomorrow's milk demand at a supermarket

5

Check a password has at least 8 characters

6

Detect a fraudulent card transaction

TOPIC 2 · HOW ML WORKS

Learn from past examples, predict on new ones

Past data
Features X and known target y

→

Training
Algorithm adjusts to match the targets

→

Model
The learned pattern

→

New data
Features only, target unknown

→

Prediction
The model's output for it

Training data

The part of the data the model learns from

Testing data

Held back; used only to check the model

The goal

Perform well on unseen data, not memorise

TOPIC 2 · EXAMPLE

Predicting a flat’s price from four features

Area (sq ft)	Location	Bedrooms	Age (years)	Price (₹ lakh) · target
650	Thane	1	12	78
900	Andheri	2	8	185
1,200	Powai	3	5	260
750	Thane	2	3	95
1,050	Andheri	2	15	190
1,000	Powai	2	2	? · predict this

The first four columns are **features**; price is the **target**. The model learns from the five complete rows, then predicts the sixth. When that flat sells, comparing prediction with reality is **evaluation**.

TOPIC 3 · TERMINOLOGY

The vocabulary for the next 18 days

Term	Meaning	Iris example (today's lab)
Dataset	Full collection of examples	150 flower records
Observation	One example (row, sample)	One flower
Feature (X)	A measurable input property	Petal length in cm
Label / target (y)	The value to predict	Species
Algorithm	The learning procedure	K-Nearest Neighbours
Model	What learning produces	A fitted KNN classifier
Parameter	Learned from the data	Slope m in y = m·x + c
Hyperparameter	Chosen before training	n_neighbors = 5
Training	Fitting the model to data	model.fit(X_train, y_train)
Inference	Using the model on new data	model.predict(X_new)

TOPIC 3 · TWO DISTINCTIONS

Algorithm vs model, parameter vs hyperparameter

Algorithm vs model

The **algorithm** is the recipe: the same for everyone.

The **model** is the cake: it depends on the ingredients, the data you used.

Same KNN algorithm + Iris data = an Iris model. Same algorithm + wine data = a different model.

Parameter vs hyperparameter

In **y = m·x + c**, the slope m and intercept c are **parameters**: the model learns them.

How many neighbours KNN looks at is a **hyperparameter**: you choose it before training.

Choosing hyperparameters well comes later in the course.

Exercises 1.2 and 1.3 · 8 min · individual

Spot the pieces, then match the terms

1.2 · Heart disease records

2,000 patients: age, blood pressure, cholesterol, smoker (yes/no), BMI, and heart disease within 5 years (yes/no).

1. Which columns are features?
2. What is the target?
3. How many observations?
4. 1,600 train: how many test?
5. What does the model output?

1.3 · Name the term

1. “The weight for area came out at 0.12”
2. “We set maximum tree depth to 4”
3. “Row 37: a 900 sq ft flat in Andheri”
4. “Whether the email is spam”
5. “Number of bedrooms”
6. “All 10,000 loan applications from 2025”

Discussion · 10 min · three groups

Find the features and the target

A · House price

Which features would you collect? What is the target? A number or a category?

B · Email spam

What features can you extract from an email? Where do the labels come from?

C · Customer segments

What describes a customer? Is there a target column? If not, what does the model produce?

Each group reports back in 2 minutes.

THEORY · HOUR 2 · 60 MIN

Types of Machine Learning

Supervised · Unsupervised · Semi-supervised and reinforcement · Classification vs regression vs clustering

TOPIC 5 · SUPERVISED LEARNING

Supervised learning learns from labelled examples

Classification

Predicts a **category**. Binary: two classes. Multiclass: three or more.

- Spam or not spam
- Disease category
- Customer will leave or stay

Regression

Predicts a **number** on a continuous scale.

- House price
- Air Quality Index
- Next month’s sales

Quick test: can the prediction be “off by 3.5”? Then it is regression. **Trap:** a 1–5 star rating or a postcode is written as a number but behaves as a category.

TOPIC 5 · ALGORITHMS

Six supervised algorithms you will meet

Algorithm	Used for	One-line intuition
Linear Regression	Regression	Fit the best straight line through the points
Logistic Regression	Classification	Estimate the probability of a class
Decision Tree	Both	A flowchart of yes/no questions
Random Forest	Both	Many trees vote
Support Vector Machine	Mainly classification	Find the widest gap between classes
K-Nearest Neighbours	Both	Look at the K most similar known examples

Names only today. Each gets its own session; in today's lab KNN is a black box.

TOPIC 6 · UNSUPERVISED LEARNING

Unsupervised learning finds structure without labels

Clustering

Group similar rows: customer segments

Dimensionality reduction

100 survey questions compressed to 3 underlying factors

Association

Items bought together: bread and butter

Anomaly detection

Rows that fit no group: unusual network traffic

Algorithms you will meet: **K-Means**, **hierarchical clustering**, **DBSCAN** and **PCA**.

TOPIC 6 · EXAMPLE

A clothing store’s customers, grouped by behaviour

Cluster	Visits / month	Average bill (₹)	Bought on discount	Name the business gives it
1	4	1,200	70%	Bargain hunters
2	1	6,500	10%	Premium occasional
3	3	3,000	30%	Loyal regulars

The algorithm finds the groups; **people name them** and judge whether they are useful. There is no single right answer to check against, which makes unsupervised models harder to evaluate.

TOPIC 7 · AWARENESS ONLY

Two more learning approaches

Semi-supervised learning

A small labelled set plus a large unlabelled set; the model uses both.

Example: 500 labelled medical scans and 50,000 unlabelled ones.

Reinforcement learning

An agent takes actions, gets rewards or penalties, and learns a strategy by trial and error.

Example: a game-playing AI, a robot learning to walk, warehouse routing.

Exercise 2.3 · 10 min · pairs

Classification, regression or clustering?

1

Telecom: will a subscriber switch to a competitor next month?

2

City: tomorrow's Air Quality Index for Mumbai

3

Retailer: group 2 lakh products with no existing categories

4

Bank: the credit limit to offer a new customer

5

School: which digit (0–9) is handwritten in a box

6

Music app: groups of listeners with similar taste

THEORY · HOUR 3 · 60 MIN

ML Workflow and Applications

The seven-stage lifecycle · Generalisation · Applications by industry · Bank fraud case study

TOPIC 9 · LIFECYCLE

Seven stages from problem to production

1. **Define the problem**
   Decision, target, success measure
2. **Collect data**
   Records with features and target
3. **Explore and prepare**
   Clean, fix and convert
4. **Split the data**
   Hold back a test set
5. **Select and train**
   Pick an algorithm, fit it
6. **Evaluate**
   Score on unseen test data
7. **Deploy and monitor**
   Use it, watch it, retrain

It is a loop, not a line. A weak evaluation sends you back to step 3 or even step 1, and monitoring tells you when to retrain. Garbage in, garbage out: a better algorithm rarely fixes bad data.

TOPIC 9 · WORKED EXAMPLE

Predicting food-delivery time, stage by stage

Stage	Food-delivery ETA (regression)
1 · Define	Predict minutes from order to doorstep; success = average error under 5 min
2 · Collect	6 months of orders: distance, prep time, hour, weather, rider load, actual time
3 · Prepare	Drop impossible times (0 or 900 min); turn “rain” into a number
4 · Split	80% of orders to train, 20% to test
5 · Train	Linear Regression as a baseline, then a Random Forest
6 · Evaluate	Test MAE = 6.2 min: not good enough, back to step 3 to add traffic
7 · Monitor	Weekly report of real vs predicted times; retrain monthly

TOPIC 9 · GENERALISATION

A model is only useful on data it has not seen

Model A

99.8% Training accuracy

71% Accuracy on new emails

Memorised the training emails; does not generalise.

Model B

94% Training accuracy

92% Accuracy on new emails

Learned real patterns; generalises well.

Which one would you deploy?

TOPIC 10 · APPLICATIONS

Where ML is used today

Industry	Application	ML type	Inputs → output
Banking	Fraud detection	Classification	Amount, place, device → fraud yes/no
Banking	Credit risk	Classification	Income, loans, history → default yes/no
Healthcare	Disease prediction	Classification	Blood tests, age → diabetic yes/no
E-commerce	Recommendations	Both	Purchase history → ranked products
Manufacturing	Predictive maintenance	Classification	Vibration, heat → failure in 7 days
Transport	Demand prediction	Regression	Time, weather, events → cab requests
Marketing	Segmentation	Clustering	Spend, frequency → customer groups
Cybersecurity	Intrusion detection	Anomaly detection	Traffic patterns → normal or suspicious

CASE STUDY · 10 MIN · GROUPS OF 4

Bank fraud detection

0.1%

of 10 lakh card transactions a day are fraud. The bank has 2 years of history, each marked fraud or genuine.

Answer as a group

1. What is the problem, in one sentence?
2. List at least six features
3. What does the model output?
4. Which type of ML is this?
5. How would you split 2 years of data for testing?
6. A model says “genuine” every time: what is its accuracy, and is it useful?
7. Which mistake is worse: blocking a genuine payment, or allowing a fraud?

THEORY · HOUR 4 · 60 MIN

Evaluating Machine Learning Models

Train, validation and test sets · Overfitting and underfitting · Classification and regression metrics · Recap quiz

TOPIC 12 · SPLITTING DATA

Three sets, three different jobs



Training

Fits the model. Like the textbook and practice problems.

Validation

Compares models and tunes settings. Like mock tests.

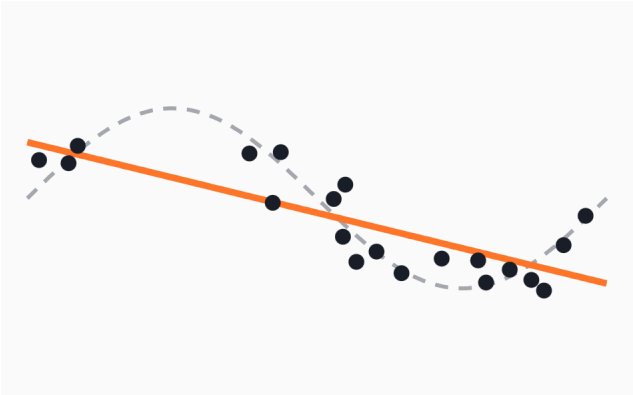
Test

Touched once, for the final honest score. Like the board exam.

Today’s lab uses train and test only. **Watch out:** shuffle sorted data, and for time-based data train on the past and test on the future.

TOPIC 13 · FIT

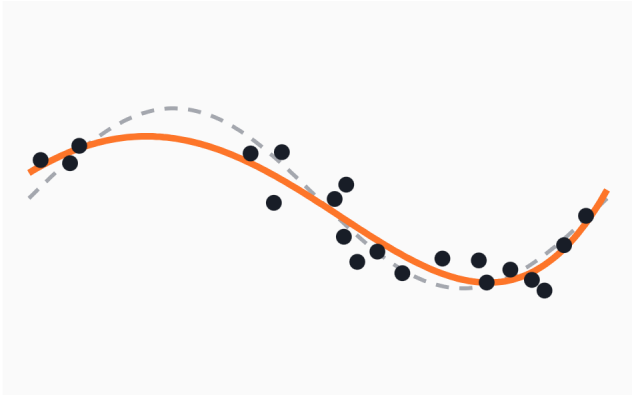
Underfitting, good fit and overfitting



Underfitting

Too simple: a straight line misses the curve

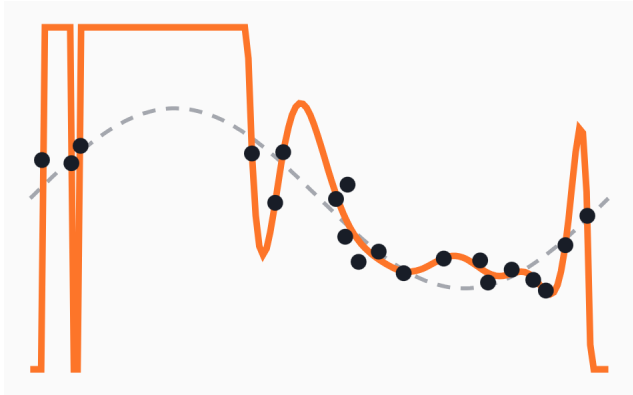
Train low · test low



Good fit

Follows the real pattern, ignores the noise

Train high · test high



Overfitting

Chases every point, swings wildly between them

Train very high · test lower

Dots: training data · orange line: the model · dashed grey line: the true pattern. Polynomials of degree 1, 4 and 15.

TOPIC 13 · DIAGNOSIS

Read the fit from training and test scores

Situation	Train score	Test score	Gap	First things to try
Underfitting	Low	Low	Small	More informative features, a more flexible model
Good fit	High	High	Small	Deploy and keep monitoring
Overfitting	Very high	Lower	Large	More data, a simpler model, regularisation

Exercise 4.1 · 4 min

Diagnose each model from its train / test accuracy: **1** 62% / 60% · **2** 100% / 68% · **3** 91% / 89% · **4** 55% / 72%

TOPIC 14 · CONFUSION MATRIX

Where a spam filter is right and wrong

	Predicted spam	Predicted not spam
Actually spam	**TP 40** Correct: spam caught	**FN 10** Error: spam missed
Actually not spam	**FP 5** Error: genuine flagged	**TN 45** Correct: genuine passed

100 test emails, 50 of them actually spam.

Diagonal = correct predictions.

Off-diagonal = the two kinds of mistake, and they rarely cost the same.

TOPIC 14 · CLASSIFICATION METRICS

Four numbers from the same confusion matrix

Metric	Question it answers	Formula	Spam example
Accuracy	Of all predictions, how many were right?	(TP + TN) / total	85 / 100 = 0.85
Precision	Of the emails flagged, how many were spam?	TP / (TP + FP)	40 / 45 = 0.89
Recall	Of all real spam, how much did we catch?	TP / (TP + FN)	40 / 50 = 0.80
F1-score	One number balancing precision and recall	2PR / (P + R)	0.84

Precision matters when false alarms are costly: a spam filter hiding a job offer.

Recall matters when misses are costly: cancer screening, fraud detection.

Accuracy misleads on imbalanced data: always “genuine” scores 99.9% and catches no fraud.

TOPIC 14 · REGRESSION METRICS

How far off are the predicted prices?

Flat	Actual	Predicted	Error	|Error|	Error²
1	200	210	+10	10	100
2	150	140	−10	10	100
3	300	330	+30	30	900
4	250	240	−10	10	100

Prices in ₹ lakh. RMSE is higher than MAE because squaring amplifies the one big miss, flat 3. For all three, lower is better.

15

MAE · average absolute error, in lakh

300

MSE · average squared error, in lakh²

17.3

RMSE · square root of MSE, in lakh

Recap quiz · 10 min · individual

Eight questions on today’s theory

1. Which is true? (a) all ML is deep learning (b) all deep learning is ML (c) ML and AI are the same (d) AI is a subset of ML

2. Traditional programming takes data and rules. ML takes data and _____.

3. Predicting next month’s electricity bill in rupees is: (a) binary (b) multiclass (c) regression (d) clustering

4. Grouping untagged news articles by topic is: (a) supervised (b) unsupervised (c) reinforcement (d) regression

5. In Iris, sepal width is a _____ and species is the _____.

6. 99% on training, 70% on test is most likely: (a) underfitting (b) overfitting (c) a good fit

7. Why do we keep a test set aside?

8. For cancer screening, prioritise precision or recall? Why?

PRACTICAL · 3 HOURS

Lab: build your first ML model in Python

```
# the pattern every scikit-learn model follows
model = KNeighborsClassifier(n_neighbors=5)
model.fit(X_train, y_train) # learn
y_pred = model.predict(X_test) # predict
accuracy_score(y_test, y_pred) # evaluate
```

Iris dataset

150 flowers

4 measurements in cm

3 species, 50 of each

Lab 1

Set up Colab, load Iris, explore and visualise

Lab 2

Split, train a KNN classifier, predict unseen flowers

Lab 3

Accuracy, confusion matrix, report, split experiments

BEFORE DAY 2

What to submit

Iris notebook

- Runs top to bottom without errors
- Exploration and two visualisations
- Model, evaluation, split experiment
- Lab tasks L1.1 to L3.4
- A 5–8 sentence conclusion

Wine assignment

- load_wine(): 178 wines, 13 features, 3 classes
- Same workflow, no step-by-step help
- Diagnose the fit
- Compare with Iris

Marked out of 20

- Runs end to end · 3
- Exploration · 3
- Visualisation · 3
- Correct workflow · 4
- Evaluation · 4
- Conclusion · 3

Conclusion question: how well does the model perform on unseen data, and what happens when the train-test split changes?

A QUESTION TO SLEEP ON

Some wine features stay below 1; another runs past 1,000. KNN compares distances. What could go wrong?

Day 2 · Data preprocessing: missing values, encoding and scaling, on your Wine notebook